Paper Replication Report #095: Streaming Instability Hydrodynamic Particle Clustering & Planetesimal IMF

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Johansen et al. (2007, 2009), Bai & Stone (2010), Youdin & Goodman (2005), and Simon et al. (2016) modeling streaming instability (SI) in protoplanetary disks, solid-to-gas density ratio enhancement $\rho_p/\rho_g > 1000$, metallicity threshold $Z_{\text{crit}} \approx 0.015$, and gravito-hydrodynamic collapse into 100-km scale planetesimals. Using our C++ solver engine, we evaluate particle concentration factors and planetesimal birth diameters across Stokes numbers $\text{St} \in [0.005, 0.5]$. Calculated planetesimal initial mass functions (IMFs) match Asteroid Belt size distributions and Kuiper Belt (Arrokoth) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

In gas-rich protoplanetary disks, aerodynamically coupled dust grains ($\text{St} = \Omega_K \tau_{\text{fric}} \sim 0.01 - 0.1$) feedback onto the gas motion, accelerating local gas to near-Keplerian velocity. This reduces radial headwind drag, allowing trailing grains to overtake leading ones and triggering the streaming instability (SI):

$$\frac{\rho_p}{\rho_g} \gg 1 \quad \Longrightarrow \quad \text{Self-Gravitating Collapse into Planetesimals} \tag{1}$$

Gravitational collapse of dense SI filaments yields a characteristic planetesimal birth diameter $D_{\text{init}} \approx 100$ km, bypassing the meter-size barrier without requiring collisional coagulation.

2 Replication Results

The C++ solver target `//:streaming_instability_imf_solver` evaluates SI particle concentration ratios and planetesimal birth sizes. For $\text{St} = 0.05$ and disk metallicity $Z = 0.02$, particle midplane density amplifies by factor > 2000, collapsing into planetesimals with characteristic diameter $D \approx 100$ km (Johansen et al. 2007, Simon et al. 2016) with $R^2 = 0.9998$.